

This disruptor is tearing down AI's memory wall

Imagine making the world's most powerful supercar engine... but no transmission could put it to use. So, its potential went untapped.

Or say you designed the world's most efficient oil rig... but even the best pipeline infrastructure couldn't move oil fast enough to keep up with it.

Bottlenecks paralyze even the most powerful systems.

But bottlenecks also create incredibly lucrative opportunities—both for the companies providing solutions, and for their investors.

That's exactly the kind of opportunity we want to tell you about today...

Over the next 12 months, the big four US-based hyperscalers alone—**Meta Platforms (META)**, **Alphabet (GOOGL)**, **Microsoft (MSFT)**, and **Amazon (AMZN)**—will spend more than \$450 billion on artificial intelligence (AI) data centers.

That's on top of the \$850 billion already spent since the AI infrastructure buildout began in late 2022—and with a few trillion dollars more to come over the next decade.

The first few years of this boom were all about companies buying as many **Nvidia (NVDA)** GPUs as fast as possible to train growing AI models.

So GPUs became the bottleneck in AI, and Nvidia became the world's most valuable company by providing them.

We've owned Nvidia in our *Disruption Investor* portfolio since September 2020. After taking profits on it twice, our position is still up over 540%.

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But the opportunity is shifting. We're entering a new phase of the AI infrastructure buildout boom, and it's minting a new class of winners.

The first AI wave was "buy GPUs."

The next wave is "feed them."

The silent bottleneck throttling AI's progress

Imagine an AI system as a kitchen...

The GPU is the chef. *Memory* technology is the pantry.

If the pantry is far away, the chef wastes time jogging back and forth to grab ingredients. But if it's right beside the stove, the chef can cook much faster.

Right now, the big bottleneck in AI is that the processors (GPUs and other AI chips) are fast but feeding them data isn't.

More than 90% of the time it takes an AI model to respond is spent fetching ingredients (data) from the far-away pantry (memory), according to leading memory company SK Hynix.

The industry calls this the "memory wall." Compute keeps getting faster, but memory hasn't kept up. So \$50,000 AI chips often sit idle, wasting electricity and time waiting for data.

Processing power has been scaling up by about 3X every two years for the last 20 years. But memory has only scaled up by about 1.4X every two years, according to research published by the Institute of Electrical and Electronics Engineers.

This gap has made memory, not compute, the primary bottleneck in AI applications.

And as AI model sizes and usage have exploded, the memory wall has become an even bigger hurdle.

For AI to continue progressing, we need better and faster ways to feed data to the processors. In other words, we need better and faster memory technology.

That's where dark horse disruptor **Rambus (RMBS)** comes in.

If we go back to our kitchen analogy, Rambus is like a crew that comes in to upgrade the pantry (memory) with features like an express cabinet attached to the stove and high-speed conveyor belts to move items from further away.

That way, the chef (the GPU) doesn't have to wait long for ingredients (data).

Without these upgrades, it doesn't matter how fast your chef can cook or how well-stocked your pantry is—everything bottlenecks as the chef waits for ingredients.

Rambus does this through two main businesses.

First, it designs and sells physical helper chips—tiny, specialized semiconductors that sit on memory modules and help data move faster and more reliably.

Memory modules are essentially plug-and-play packages used in computers and data center servers.

They're comprised of a circuit board, DRAM chips—from companies like **Micron Technology (MU)**, SK Hynix, and Samsung—and supporting chips (like Rambus's) that make them work and boost their performance.

DRAM—short for dynamic random-access memory—are the workhorse memory chips in data centers that temporarily hold data (programs and files) in active use for processors (GPUs, CPUs, etc.) to grab quickly.

Second, Rambus licenses its technology blueprints (called “intellectual property” or “IP”) to other companies, including Nvidia, Micron, SK Hynix, Samsung, **Intel Corp. (INTC)**, **Broadcom (AVGO)**, and **Qualcomm (QCOM)**.

Rambus then collects royalties when these companies use its inventions in their own products.

It's a powerful combo business model that generated \$678.5 million in revenue over the past year, up 32.6% from the year before.

3 ways Rambus is solving the memory wall problem

Remember, the memory wall is like the speed limit slowing down AI. Today's AI processors are insanely fast, but they often sit idle waiting for data to arrive.

Rambus has come up with three unique solutions to help solve that bottleneck...

Solution #1: Enabling super-fast memory for AI chips (HBM)

The most critical bottleneck in AI happens at the processor (GPU) level.

In older AI chips, multiple DRAM memory chips were positioned around the central processor on the circuit board. It's like breaking up the pantry into several cabinets and putting them far away from the chef.

Data had to travel along relatively long wires on the board between the GPU and these external memory chips.

In new AI chips, this paradigm has changed. Memory is now stacked right next to or on top of the processor in the same package, eliminating the long travel distance.

The result of this integration is High-Bandwidth Memory (HBM).

HBM takes several DRAM memory chips and stacks them vertically—essentially forming a 3D memory cube—and connects them directly to the processor with tiny wires called through-silicon vias rather than through a traditional circuit board path.

Instead of data traveling centimeters across a circuit board through maybe 64 tiny wires, HBM uses over 1,000 wires spanning just millimeters. It's like replacing a few long and narrow garden hoses with many short, massive fire hoses.

Or to stick with our kitchen/chef/pantry analogy: It's like stocking an express pantry cabinet with the chef's most-used ingredients and attaching it directly to the stove.

With HBM, the distance for data to travel is vastly reduced and the width of the "pipe" between the processor and memory is vastly increased. This means multiple-fold increases in the amount of data that can be fed to the processor per second.

Here's where Rambus comes in:

To use HBM, an AI chip needs a specialized "traffic brain" (called an HBM controller) to manage all those connections and ensure data flows correctly.

Rambus designs these controllers and licenses the blueprints to chip makers. Its latest HBM4 controller—the industry's first—can move 2.5 terabytes of data per second. That's equivalent to about 300 HD movies every second.

More than 100 different chips today are using Rambus's HBM controller technology—including the HBM stacks manufactured by Micron, SK Hynix, and Samsung that you'll find in Nvidia's and Alphabet's most advanced AI chips.

Importantly, Rambus is consistently first to market with each new generation of "traffic brain" chips, making it the preferred provider of HBM controller technology to solve big tech's memory bandwidth problems.

Solution #2: Expanding memory capacity (CXL)

The second bottleneck occurs at the system level.

Today's AI models are so large, they often don't fit in the memory attached to a single processor. It's like having a recipe that requires more ingredients than can fit in your kitchen.

CXL (Compute Express Link) is a new technology that lets multiple processors share a common pool of memory they can access as needed. It's like multiple kitchens sharing a giant communal pantry.

Rambus's CXL controller IP, which the company licenses to chipmakers, plays a critical role in enabling this sharing.

It matters for AI because training and inference love memory, and CXL makes memory more scalable and flexible. That's a big deal when model sizes and job sizes constantly change.

Big tech companies like Alphabet, Microsoft, and Amazon are starting to deploy CXL in their AI data centers—with significant ramping expected over the next few years—because it lets them use their expensive memory assets more efficiently and cut costs.

For example, instead equipping 10 servers with 10 terabytes of dedicated memory, they might only need to buy half that to achieve the same performance using a CXL shared pool those servers can access.

Solution #3: Speeding up server memory (DDR5 and MRDIMM)

The third bottleneck involves the main memory used in servers—called DDR (Double Data Rate) memory. It's like the main room-temperature pantry.

DDR is DRAM that's gone through a series of enhancements to give it more bandwidth, improved power efficiency, increased memory density, and better reliability and scalability.

DDR5 is the newest generation, about twice as fast as DDR4. But even DDR5 still isn't fast enough for cutting-edge AI models and high-performance computing applications.

So Rambus (in collaboration with the big industry players like Micron, SK Hynix, and Samsung) invented something clever called MRDIMM (Multiplexed Rank Dual Inline Memory Module).

Here's how it works: Traditional memory modules are like a single-lane road where cars (data) must travel one behind the other. MRDIMMs create two lanes and interweave the traffic, effectively doubling the bandwidth without needing new memory chips.

MRDIMMs need specialized helper chips (called memory interface chips) to make these modules run fast and reliably. Rambus supplies these chips.

This is brilliant because it means data centers can effectively double their memory bandwidth simply by incorporating Rambus's specialized memory interface chips, which are added to the modules during manufacturing. And this upgrade is plug-and-play, compatible with existing DDR5 server architectures and DIMM slots.

What's more, each MRDIMM memory module needs 11 Rambus chips, compared to just one for standard modules—essentially 11Xing Rambus's revenue opportunity for each module sold.

We should note that MRDIMM is brand-new technology. Rambus CEO Luc Seraphin says:

MRDIMM is staged to enter the market toward the end of 2026, depending on the availability of platforms. This is not the next-generation platform, but the one after. But it's important to engage with customers very early on.

We believe each of these three cutting-edge memory solutions will make Rambus mission-critical in Phase 2 of the AI buildout.

Big tech companies will increasingly turn to Rambus to solve their memory wall pain points, which means it can expect robust revenue growth for years to come.

How Rambus makes money “feeding the GPUs”

Rambus makes money three ways:

1. **Product revenue (52.3% of revenue in Q3 2025):** These are the physical memory interface (helper) chips that sit on the modules. They let servers run faster memory modules with more bandwidth per slot—cutting the amount of time processors must wait for data from memory.

In the nine months ending September 30, 2025, this business grew 44.7% year over year to a record \$251.0 million as data centers upgrade to DDR5 (and start incorporating MRDIMM toward the end of next year).

2. **Royalties (36.5% of revenue in Q3 2025):** When Micron makes a memory chip or Nvidia builds a GPU, they often pay Rambus regular royalties to license and use its patented technologies for HBM and other purposes.

Rambus's 2,000-plus issued patents provide recurring, high-margin royalty revenue. In the nine months ending September 30, 2025, this revenue stream grew 23.7% year over year to \$207.7 million.

3. **Contract and other revenue (11.2% of revenue in Q3 2025):** This revenue mostly comes from upfront fees customers pay to license Rambus's IP and services the company provides to help customers customize their products, like tweaking memory tech for a specific AI server. In the nine months ending September 30, 2025, this revenue stream grew 8.5% year over year to \$58.7 million.

Meanwhile, it's important to point out that Rambus has extended patent licensing agreements with the “Big Three” memory manufacturers:

- Micron Technology, through 2029
- SK Hynix, through 2034
- Samsung Electronics, through 2033

Total revenue for the nine months ending September 30, 2025 reached \$517.4 million, up 30.8% year over year, thanks to accelerating demand across Rambus's business segments—especially products.

Put simply, Rambus is already benefiting handsomely from the global shift in AI infrastructure spending from “buying GPUs” to “feeding them.” And we expect those benefits to compound over the coming years.

Rambus's path to profits

Rambus is already profitable, generating \$228.8 million in net income over the past year, which reflected growth of 29.9%.

Meanwhile, operating cash flow and free cash flow both hit new records this past year of \$319.2 million and \$292.4 million, respectively. Ample cash generation is critical for Rambus to reinvest in its memory designs and maintain technological leadership.

As we alluded to earlier, AI data centers are still transitioning from DDR4 to the newer, faster DDR5 memory for their servers. Rambus estimates it will generate multiple billions of dollars in DDR5 chip revenue over the course of the product cycle.

The *newest* AI data centers are using both DDR5 and HBM, because they play complementary roles depending on the specific workload, hardware, and system architecture. We expect Rambus's HBM controller revenue to shoot higher with the continued proliferation of AI.

Bottom line: Rambus holds a unique—and lucrative—position in the AI infrastructure landscape.

The memory wall isn't some theoretical future problem. It's the biggest bottleneck limiting AI performance today. Every AI company—from OpenAI to Alphabet to all the startups you read about—face this challenge. And they can't solve it with software or clever algorithms.

They need hardware solutions, and that's exactly what Rambus provides—the hardware solutions solving the bottleneck to keep the world's fastest chips fed with data.

Rambus has spent 35 years becoming the world's expert at moving data between memory and processors. Now, as AI creates unprecedented demand for memory bandwidth, it's perfectly positioned for exploding revenue and profits during Phase 2 of the AI infrastructure buildout boom thanks to its decades of specialized expertise.

Rambus is on track to generate about \$707 million in total revenue for the full year of 2025, which would reflect growth of 27%.

That's the slowest annual growth we expect to see over our anticipated holding period of three to five years.

Going forward, we think Rambus can grow revenue at a compound annual rate of 30% to 35% for the next few years. Using the midpoint of that forecasted range, we calculate a GARP Quotient of 0.45.

That's near the upper limit of (but still within) the "Great" tier we talked about in [The Growth Stock Anomaly special report](#). So the stock is still attractively priced given our growth expectations. It's just not as cheap as many of our other positions at the time of entry.

Action to take: Buy Rambus (RMBS) up to \$110 and set a 65% hard stop.

4 fast-growing crypto miners pivoting to AI

Back in May, we [recommended Digi Power X \(DGXX\)](#) because the company—like **CoreWeave (CRWV)** before it—is leveraging the assets it acquired for crypto mining to pivot into AI.

The business model is simple: Instead of companies buying and setting up their own infrastructure to run AI workloads—like GPU servers, cooling systems, and electricity—Digi puts it all together to rent out "slices" of computing power to customers who need it for AI.

Digi and CoreWeave are part of the fast-growing "neocloud" trend. These companies typically provide faster access, lower prices, and more flexible options than traditional big cloud providers like Amazon Web Services, Microsoft Azure, or Google Cloud.

The table below shows four other fast-growing crypto miners leveraging their assets to become neocloud providers for AI. While we're not recommending any of them today, they're interesting stocks that are on our radar and worth knowing about.

As a reminder, the stocks we target have a market cap between \$200 million and \$20 billion. These stocks trade on American exchanges, but they aren't necessarily American companies.

IREN Ltd. (IREN), the largest name on our list with a market cap north of \$10 billion, is aggressively pivoting into AI. The company is scaling up its "IREN AI Cloud" business with GPU clusters for training and running AI models, powered by renewable energy—mostly hydro, wind, and solar.

As a Nvidia Preferred Cloud Partner, IREN gets priority access to its GPUs. So while others wait months to get their hands on these chips, IREN's neocloud customers can benefit from its early access.

Nvidia's gold seal of approval makes IREN attractive to big enterprises looking to expand their AI stack. In November, for example, IREN inked a five-year, \$9.7 billion deal providing Microsoft with AI cloud infrastructure.

So IREN is scaling super-fast. Over the past year, it generated revenue of \$685.3 million, which reflected annual growth of 233%.

Bit Digital (BTBT) is pivoting into AI cloud solutions through its WhiteFiber high-performance computing division. But rather than going “all in” on AI like IREN, Bit Digital is also reinventing itself as an **Ethereum (ETH)** staking company—generating revenue by “validating” Ethereum.

So Bit Digital is more of a hybrid miner/cloud operator than most of its neocloud peers. This dual focus is a way to hedge its bets. But the strategy sacrifices significant near-term growth potential for the added perceived safety of diversification.

You can see this playing out in Bit Digital’s financial results. In its most recent quarter, for example, total revenue grew 33% year over year as a 48% increase in cloud service revenue helped offset a 27% decline in mining revenue.

CleanSpark (CLSK) has been around since 1987. It started as a company called SmartData and was originally focused on alternative fuel tech and energy-related R&D. Then, in 2016 following a couple acquisitions, the company started focusing on microgrids powered by “clean” energy sources like solar and wind.

It leveraged its energy tech to become an ultra-efficient **bitcoin (BTC)** miner in 2021, with six mining data centers across Georgia, Mississippi, Tennessee, and Wyoming. The company is now working on a pivot to AI.

CleanSpark is a later entrant to the neocloud space. It hasn’t generated revenue from AI yet, but it’s moving quickly. In October 2025, the company acquired a 271-acre site outside Houston for its first dedicated AI data center. The company also formed a strategic partnership with Submer in October to design and construct efficient AI data centers across North America.

CleanSpark ended last quarter with \$1.2 billion in bitcoin on hand. And if the company can leverage its power-efficient scaling in the neocloud space, it could start to attract lucrative AI deals like IREN and the next stock on our list...

TeraWulf (WULF) was purpose-built for “clean,” sustainable bitcoin mining by experienced energy infrastructure entrepreneurs. It’s now leveraging its power infrastructure, land, grid connections, and clean energy assets to pivot into AI. And the company’s already inked some big neocloud deals.

Google-backed Fluidstack, for example, has leased out several billion dollars’ worth of TeraWulf’s capacity over the next decade. And the company says it has now signed more than \$17 billion in long-term neocloud contracts with customers, providing a solid foundation for future growth.

4 fast-growing crypto miners pivoting to AI							
Rank	Ticker	Name	Description	Market Cap (\$M)	Stock Price	Weighted Growth Score	GARP Quotient
1	IREN	IREN Ltd.	AI cloud, some bitcoin mining	\$10,148	\$35.80	131.04	0.06
2	BTBT	Bit Digital, Inc.	Ethereum staking plus AI cloud	\$654	\$2.02	88.88	0.04
3	CLSK	CleanSpark, Inc.	Bitcoin miner adding AI exposure	\$2,863	\$11.20	79.83	0.03
4	WULF	TeraWulf, Inc.	AI data centers powered by nuclear	\$4,936	\$11.79	62.62	0.24

For the most up-to-date guidance on our current positions, [click here](#) to view the portfolio online.

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